

Topic 2.6 - Competing Function Model Validation

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Select plausible linear, quadratic, or exponential models from patterns, context, and residual behavior.
- Compute and interpret signed residuals and absolute error measures.
- Validate a model within a domain and articulate consequences of overprediction, underprediction, and extrapolation.

Section A: Foundation and representation

Question 1

Core skill

No calculator

Classify each exact pattern from consecutive outputs: (a) 7, 12, 17, 22; (b) 3, 8, 15, 24; (c) 5, 10, 20, 40.

Question 2

Core skill

No calculator

For observed $y = 42$ and predicted $\hat{y} = 38.7$, compute the residual and state whether the model overpredicts or underpredicts.

Question 3

Core skill

No calculator

Observed values are 12, 18, 25, 33 and Model A predicts 13, 17, 27, 30. Find all residuals and the MAE.

Question 4

Core skill

No calculator

Model B predicts 11, 20, 24, 34 for the same observed values 12, 18, 25, 33. Compare its MAE with Model A from the previous question.

Question 5

Residual analysis

Calculator permitted

The table shows observed values and predictions.

x	0	1	2	3	4
y	80	64	51	41	33
$\hat{y}_E$	80	64	51.2	40.96	32.768
$\hat{y}_L$	80	68	56	44	32

Compute residuals for both models and select the stronger model.

Question 6

Residual pattern

No calculator

A residual plot for a model shows points steadily rising from about -12 to $+15$ as x increases. What does the pattern indicate?

Section B: Application, comparison, and error analysis**Question 7**

Validation

No calculator

A model with the smallest training error performs poorly on new data. Name two likely explanations.

Question 8

Consequences

No calculator

A hospital demand model has residuals mostly positive during peak hours. Explain the practical risk.

Question 9

Model structure

No calculator

Data for a cooling object are positive and approach room temperature, but an unshifted exponential regression approaches 0. Explain why a shifted exponential model may be more defensible.

Question 10

Context

No calculator

A quadratic model and an exponential model have nearly equal MAE on $0 \leq x \leq 6$. The context is compound interest. Which model is more defensible and why?

Question 11

Extrapolation

No calculator

A regression is fitted using data for ages 10–18. A report uses it to predict age 45. Evaluate the prediction.

Question 12

Model design

No calculator

Construct a residual-based comparison rule that gives extra weight to underprediction when shortages are costly.

Section C: AP-style multiple choice**MCQ 1**

No calculator

Using $r = y - \hat{y}$, a residual of -6 means:

- A. The model underpredicts by 6.
- B. The model overpredicts by 6.
- C. The observed value is negative.
- D. The model is invalid.

MCQ 2

No calculator

Which residual plot most supports a model?

- A. Residuals forming a clear U-shape
- B. Residuals increasing steadily
- C. Residuals scattered around zero with no clear pattern
- D. All residuals positive

MCQ 3

No calculator

Constant second differences are evidence for which exact model family?

- A. Linear
- B. Quadratic
- C. Exponential
- D. Logarithmic

MCQ 4

No calculator

A model has lower MAE but contradicts the known physical mechanism. What is the best conclusion?

- A. Always choose the lower MAE.
- B. Discard context and keep the regression.
- C. Consider both fit and mechanism; the lower MAE alone is not decisive.
- D. Average the two models.

Section D: Free-response reasoning

FRQ 1 - Competing models and residuals

Calculator permitted

6 points

The observed outputs for $x = 0, 1, 2, 3, 4$ are 50, 61, 74, 90, 109. Model L predicts 50, 64, 78, 92, 106. Model E predicts 50, 60, 72, 86.4, 103.68. (a) Compute the residuals for both models. (b) Compute each MAE. (c) Select the better model on this interval and justify using both size and pattern. (d) State one reason the conclusion may change with additional data.

FRQ 2 - Validation in context

Calculator permitted

6 points

A school uses $D(t) = 1200(1.06)^t$ to predict device demand t years after 2026. Actual demand for $t = 0, 1, 2, 3$ is 1200, 1265, 1348, 1460. (a) Compute predictions and residuals. (b) Describe the residual pattern. (c) Evaluate the model for short-term planning. (d) Explain why using the model at $t = 20$ could be risky and name one alternative validation step.
